The spectral pentagon interface: parameter continuity and regular Weyl representations

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Abstract

We examine two possible explanations of a previously certified nonzero pentagon residual: an exceptional deformation parameter and a different regular representation of the same Weyl relations. Dominated convergence and the standard Stone–von Neumann theorem exclude these explanations within the specified bounded spectral construction. These are applications of existing mathematics, conditional on the earlier computer-assisted bound; they do not independently certify that bound. No quantitative parameter interval or finite-G physical correlator is obtained.

1 Fixed input and literature interface

All variables are dimensionless. Write $t > 0$ for the deformation parameter denoted $\hbar$ in Kim–Scarinci (KS) [1]; it is distinct from the fixed CCR normalization π. Let

$$\mathcal{H} = L^2(\mathbb{R}^2, dx\,dy/(2\pi^2)), \quad P_j = -\pi i\partial_j, \quad M = e^{iXY/\pi}.$$

The KS compact-product formula and conjugate pairing give, almost everywhere,

$$F_t(x,y) = \prod_{j=0}^{\infty} \frac{1+e^{x-(2j+1)\pi t-ity}}{1+e^{x-(2j+1)\pi t+ity}} \frac{1+e^{y-(2j+1)\pi/t-ix/t}}{1+e^{y-(2j+1)\pi/t+ix/t}}. \tag{1}$$

Assign any modulus-one value at a zero denominator. Define by ordinary joint spectral calculus

$$A_t = F_t(X,Y), \quad B_t = F_t(P_y,P_x), \quad D_t = A_tB_t - B_tM^{-1}B_tMA_t.$$

The normalized input is $f(x,y) = \sqrt{2}e^{-(x^2+y^2)/(2\pi)}$. Our preceding certificate [4] gives

$$\|D_1 f\| > b, \qquad b = 0.04489. \tag{2}$$

Equation (2) is reused, not recomputed here. KS Proposition 6.15 states the pentagon for self-adjoint generators satisfying these Weyl relations. Its $\Lambda = 1$ proof refers back to parameter continuation. Kim's Proposition 7.4 [2] explicitly cites the same argument; it is not an independent check. The following implications concern exactly (1) and these bounded operators.

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2 The residual is not confined to the self-dual parameter

Proposition 1. *The maps $t \mapsto A_t$, $t \mapsto B_t$, and $t \mapsto D_t$ are strongly continuous on $(0,\infty)$. Consequently there is an $\varepsilon > 0$ such that*

$$|t-1| < \varepsilon \quad \Longrightarrow \quad \|D_t f\| > b/2 = 0.022445. \tag{3}$$

The value of ε is not estimated here.

Proof. Fix $t_0 > 0$ and a compact interval $[a,c] \subset (0,\infty)$ around it. For $|x|, |y| \le R$ the absolute values of the small terms in the two denominators of (1) are bounded by

$$e^{R-(2j+1)\pi a}, \qquad e^{R-(2j+1)\pi/c}.$$

Both sequences are summable, uniformly in $t \in [a,c]$. For large j these bounds are below $1/2$; the corresponding ratios differ from one by at most four times these bounds. Hence the infinite tails converge uniformly on these compact parameter and spatial sets.

At $t = t_0$, every zero denominator lies in the countable set

$$E_{t_0} = \{((2m+1)\pi t_0, (2n+1)\pi/t_0) : m, n \in \mathbb{Z}\}.$$

This set may overcount cancellations, which is harmless. For a fixed $(x,y) \notin E_{t_0}$, finitely many initial denominators stay nonzero in a parameter neighborhood, while the tails already have a uniform nonzero lower bound. Thus $F_t(x,y) \to F_{t_0}(x,y)$ almost everywhere as $t \to t_0$. Also $|F_t| = 1$ almost everywhere. For every $g \in \mathcal{H}$, dominated convergence therefore gives

$$\|(A_t - A_{t_0})g\|^2 = \int |F_t - F_{t_0}|^2 |g|^2 \, d\mu \longrightarrow 0, \qquad |F_t - F_{t_0}|^2 |g|^2 \le 4|g|^2.$$

The fixed Fourier transform diagonalizes (P_y, P_x), so it gives the same conclusion for B_t. Products of strongly continuous uniformly bounded families are strongly continuous: for example,

$$\|(U_t V_t - U_0 V_0)g\| \le \|U_t\| \|(V_t - V_0)g\| + \|(U_t - U_0)V_0 g\|.$$

Apply this to the two words defining D_t; M is fixed and unitary. Finally use (2) and continuity of $D_t f$ to obtain (3). □

This is strong continuity for a real parameter, not operator-norm continuity, differentiability, or holomorphic continuation. It includes both rational and irrational t near one. KS Proposition 6.6(SB1) distinguishes real Barnes parameter h from $h = it$; its remark on rational real h cannot be applied to $t = 1$ as an exclusion.

3 Changing a regular realization cannot remove the word

Let (Q_1, Q_2, R_1, R_2) be self-adjoint on a nonzero separable Hilbert space $\mathcal{H}_\pi$, with the full Weyl versions of

$$[Q_1, R_2] = [Q_2, R_1] = \pi i, \quad [Q_1, R_1] = [Q_2, R_2] = [Q_1, Q_2] = [R_1, R_2] = 0.$$

In particular every individual exponential group is strongly continuous. After the relabeling $q = (Q_1, Q_2)$, $p = (R_2, R_1)/\pi$, these are the ordinary nondegenerate two-degree-of-freedom CCR. The regularity and finite-dimensional hypotheses of the Stone–von Neumann theorem are exactly those in Dereziński [3], §3.1–3.2, Theorem 4.

Proposition 2. *Use the same Borel function F_t and ordinary joint spectral calculus to define the pentagon word D_t^π in this representation. There are a nonzero multiplicity space $\mathcal{K}$ and a unitary $U : \mathcal{H} \otimes \mathcal{K} \to \mathcal{H}_\pi$ such that*

$$U^{-1} D_t^\pi U = D_t \otimes I_{\mathcal{K}}. \tag{4}$$

Thus $\|D_1^\pi\| > b$, and $\|D_t^\pi\| > b/2$ in the neighborhood above.

Proof. The cited theorem supplies the intertwiner for all Weyl unitaries and hence their self-adjoint generators and spectral measures. The constant in the Lebesgue measure of $\mathcal{H}$ is absorbed in a unitary normalization. Joint Borel functional calculus commutes with this equivalence. For the middle factor the closures of the sums satisfy

$$(X + P_y, Y + P_x) = M^{-1}(P_y, P_x)M,$$

as follows first on the Schwartz core by differentiating the chirp, then by unitary conjugation. In particular the sums strongly commute, and $F_t(X + P_y, Y + P_x) = M^{-1} B_t M$. This transports every factor of the pentagon and proves (4). For any unit vector $k \in \mathcal{K}$,

$$\|D_t^\pi U(f \otimes k)\| = \|D_t f\|,$$

which proves the claimed lower bounds. □

This directly applies the standard multiplicity theorem; it is not a new representation-classification result. A simultaneous symplectic change of the four generators is included. Multiplicity, normalization of Lebesgue measure, or a mere choice of coordinate polarization does not change the conclusion. A representation with a different spectrum, a nonregular Weyl action, or only formal commutators is outside the stated hypothesis and must be identified explicitly before comparison.

4 What this closes, and what remains open

Dense-domain changes. All factors in the final word are bounded. If the same two bounded words agree on a norm-dense subspace, continuity extends the equality to the full Hilbert space. Thus a different dense test domain cannot repair this particular bounded identity. It can still matter for intermediate unbounded factors or for a construction defining different bounded operators.

Null sets and states. In the canonical realization each paired joint spectral measure is unitarily related to Lebesgue measure. Changing F_t only on its countable exceptional set has no effect, also after (4). Some individual states could nevertheless lie in $\ker D_t$; our conclusion is the failure of an operator identity, not a claim that every state's residual is nonzero. A proposed constrained physical sector requires its own observable algebra and inner product.

Evidence boundary and next physical question. The positive input (2) remains computer-assisted. This note neither independently re-integrates its overlaps nor formally verifies the numerical library. It derives consequences of that input and identifies two inadequate explanations of the discrepancy with KS's literal spectral statement. It does not rule out every quantization of three-dimensional gravity or any complete de Sitter duality.

The next physical task is to compare an actual observer state's reflection and gluing prescription with the local algebra used here. A replacement must specify the changed spectrum, scalar function or functional calculus, and map its observables and inner product to that physical problem. It is insufficient to give another formula with the same name. The finite-G observer and full holographic dictionary remain open; routine refinement of the previous FFT is not required by either proposition in this note.

Reading and reproducibility. On 5 November 2026 the arXiv records still list KS v2 (21 June 2024) and Kim v2 (5 July 2023). A targeted search and the checked author and publisher pages did not locate a relevant correction; this is not an exhaustive absence claim. The source ledger records exact reading locations. The previous bound and its file hashes are checked by the accompanying input audit, which is an integrity check, not another proof of the overlap enclosures. The present note needs no parameter scan. Source, PDF, build log, and audit hashes are retained separately.

References

[1] H. K. Kim and C. Scarinci, *A quantization of moduli spaces of 3-dimensional gravity*, arXiv:2112.13329v2, §6.3–6.5, Propositions 6.6, 6.11, 6.15.

[2] H. K. Kim, *Three-dimensional quantum gravity from the quantum pseudo-Kähler plane*, arXiv:2112.13962v2, §5, Propositions 7.3–7.4.

[3] J. Dereziński, *Introduction to representations of the canonical commutation and anticommutation relations*, arXiv:math-ph/0511030v2, §3.1–3.2, Theorem 4, p. 12.

[4] A. Day and Chengheng, *A posteriori certification of a bounded pentagon word: Gaussian moments and validated overlap integrals*, project working note, 1 November 2026; `manuscript/residual-certificate/main.tex` and associated evidence.